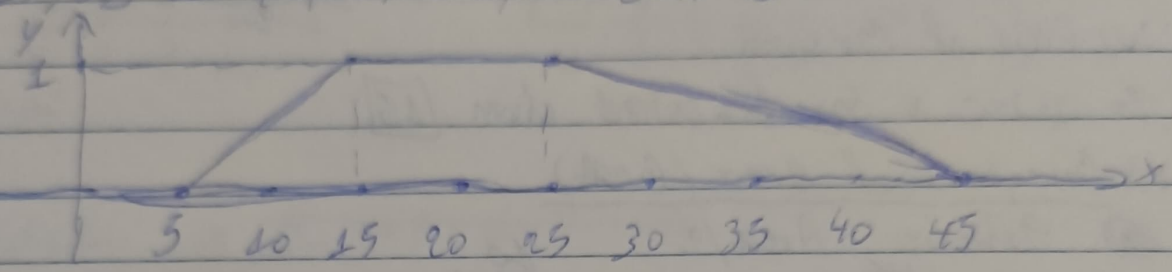
$\boxed{3.6}$ H [illegible]

a)

$$\mu_A(x) = \begin{cases} 0, & x \le 5 \\ (x-5)/20, & 5 \le x \le 15 \\ 1, & 15 \le x \le 25 \\ (45-x)/20, & 25 \le x \le 45 \\ 0, & x \ge 45 \end{cases}$$

โดยจุดที่ [illegible] ที่ช่วง [5, 15], [illegible] ช่วง [15, 25] และ [illegible] ช่วง [25, 45]



i) Center of Area Method (COA)

$$x_0 = \frac{\int_5^{45} x \cdot \mu_A(x)\,dx}{\int_5^{45} \mu_A(x)\,dx} \qquad [5, 45] \text{ support}$$

$$\int_5^{45} x\mu_A(x)\,dx = \int_5^{15} x \cdot \frac{x-5}{20}\,dx + \int_{15}^{25} x \cdot 1\,dx + \int_{25}^{45} x \cdot \frac{45-x}{20}\,dx =$$

$$= \frac{1}{20}\left[\frac{x^3}{3} - \frac{5x^2}{2}\right]_5^{15} + \left[\frac{x^2}{2}\right]_{15}^{25} + \frac{1}{20}\left[\frac{45x^2}{2} - \frac{x^3}{3}\right]_{25}^{45}$$

$$\frac{1}{20}\left(\left(\frac{3375}{3} - \frac{1125}{2}\right) - \left(\frac{125}{3} - \frac{125}{2}\right)\right) = \cdots = 58.\overline{3}$$

$$\frac{1}{2}(25^2 - 15^2) = 200$$

$$\frac{1}{20}\left(\frac{45 \cdot 2025}{2} - \frac{91125}{3} - \left(\frac{45 \cdot 625}{2} - \frac{15625}{3}\right)\right) = \cdots \approx 319.16$$

$$\int_5^{15} \frac{x-5}{20}\,dx + \int_{15}^{25} 1\,dx + \int_{25}^{45} \frac{45-x}{20}\,dx$$

$$\frac{1}{20}\left[\frac{(x-5)^2}{2}\right]_5^{15} = 5 \qquad 10 \qquad \frac{1}{20}\left[\frac{(45-x)^2}{2}\right]_{25}^{45} = \cdots = 10$$

ดังนั้น $x_0 = \dfrac{58.\overline{3} + 200 + 319.16}{5 + 10 + 10} = \boxed{23.096}$

ii) Height-center of area method

~~It gives the same result because~~

~~Since we have symmetry~~

Όπου $a=0.5$ έχει $\dfrac{\int_{10}^{32.5} x\mu_A(x)\,dx}{\int_{10}^{32.5} \mu_A(x)\,dx}$

Αλλά οι τιμές θα κυμαίνονται από 0 ή 0.5. Άρα μηδενίζεται το $\int_{40}^{50}$

και τελικά αλλά ακριβής παρεμβαλλόμενη που υπερβαίνει κάποιες έχει

$\boxed{x_0 = 23.096}$

iii) Max Criterion

μέσος όρος απ'ους μέγιστες τιμές: $\dfrac{15+25}{2} = \boxed{20}$

iv) First of Maxima

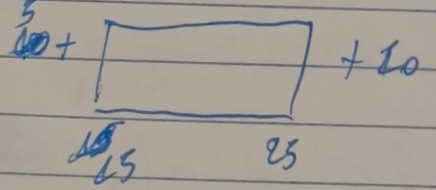
Το πρώτο x όπου $\mu_A(x)=1$ είναι $\boxed{15}$

v) Bisector of Area (BOA)

Ψάχνουμε Area/2

$\text{Area} = \int_5^{45} \mu_A(x)\,dx = \dfrac{10\cdot 1}{2} + 10\cdot 1 + \dfrac{20\cdot 1}{2} = \boxed{25}$

Αρκεί να βρω x για A=12.5



~~$5+x = 25-x+10 \Leftrightarrow 2x=30 \Leftrightarrow x=15$~~

$5+x-15 = 25-x+10 \Leftrightarrow 2x=45 \Leftrightarrow \boxed{x=22.5}$

β) Αν το σύνολο αναφοράς (universe of discourse) περιέχει τιμές του ακέραιους απ'το 0 μέχρι το 100 τότε:

i) Center of Area (COA): $x_0 = \dfrac{\sum_{i=1}^{n} x_i \mu_A(x_i)}{\sum_{i=1}^{n} \mu_A(x_i)}$

$(5=0)$ $\dfrac{\sum_{i=6}^{15} x_i \frac{x_i-5}{10} + \sum_{i=16}^{25} x_i\, 1 + \sum_{i=26}^{45} x_i \cdot \frac{45-x_i}{20}}{\sum_{i=6}^{15} \frac{x_i-5}{10} + \sum_{i=16}^{25} 1 + \sum_{i=26}^{44} \frac{45-x_i}{20}}$ $(45==0)$

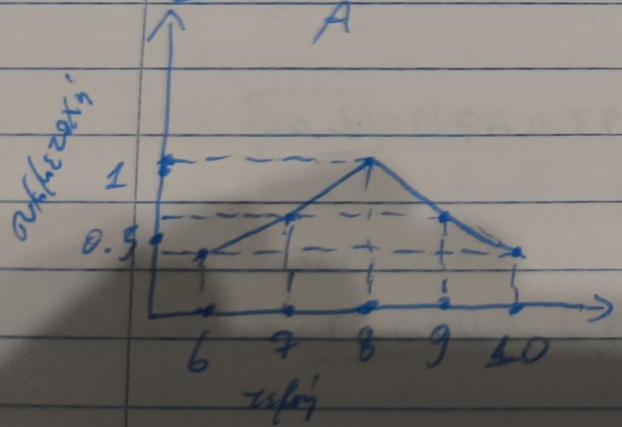
Αυτοί οι τύποι αξιοποιούν παραλλαγές το 1ου ~~[illegible]~~ ενώ το BOA αποφυλάσσεται στο $\boxed{22}$

Άσκηση 3.1

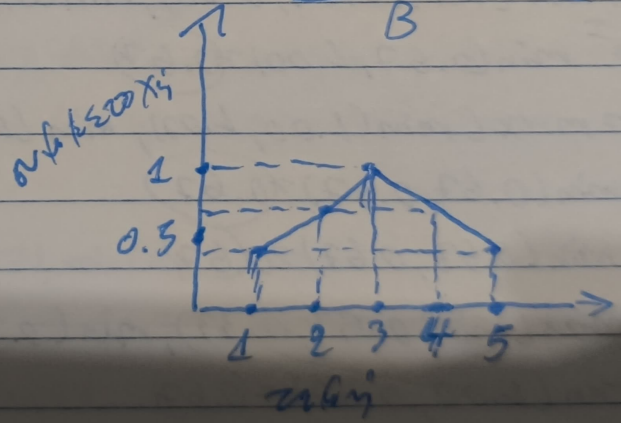
(αβέβαιη τιμή κοντά στα 8 και 3 αντίστοιχα)

i) Είναι ~~ο~~ ασαφείς αποδεκτοί αριθμοί, το A για τον αριθμό 8, ενώ το B για τον αριθμό 3. Παρατηρείστε ότι είναι ίδια κατανομή

Ασαφές 8



Ασαφές 3



ii) Eşitliğinde her zaman aralığı zaman zaman kullanılan formülü: $\mu_{A\cdot B}(z) = \max_{z=x\cdot y} \min(\mu_A(x), \mu_B(y))$

$\mu_{C=A\cdot B}(6) = \min(\mu_A(6), \mu_B(1)) = \min(0.33, 0.33) = \underline{0.33}$

$\mu_{C=A\cdot B}(12) = \min(0.33, 0.67) = \underline{0.33}$

$\mu_{C=A\cdot B}(16) = \min(0.67, 0.67) = 0.67$

$\mu_{C=A\cdot B}(18) = \min(1.00, 0.67) = 0.67$

$\mu_{C=A\cdot B}(18) = \max(\min(0.33, 1.00), \min(0.67, 0.67)) = 0.67$

$\mu_{C=A\cdot B}(24) = \min(0.67, 1.00) = 0.67$

$\mu_{C=A\cdot B}(24) = \max(\min(1.00, 1.00), \min(0.33, 0.67)) = \boxed{1.00}$

$\mu_{C=A\cdot B}(27) = \min(0.67, 0.67) = 0.67$

$\mu_{C=A\cdot B}(28) = \min(0.67, 0.67) = 0.67$

$\mu_{C=A\cdot B}(30) = \max(\min(0.33, 0.67), \min(0.33, 1.00)) = \underline{0.33}$

$\mu_{C=A\cdot B}(32) = \min(1.00, 0.67) = 0.67$

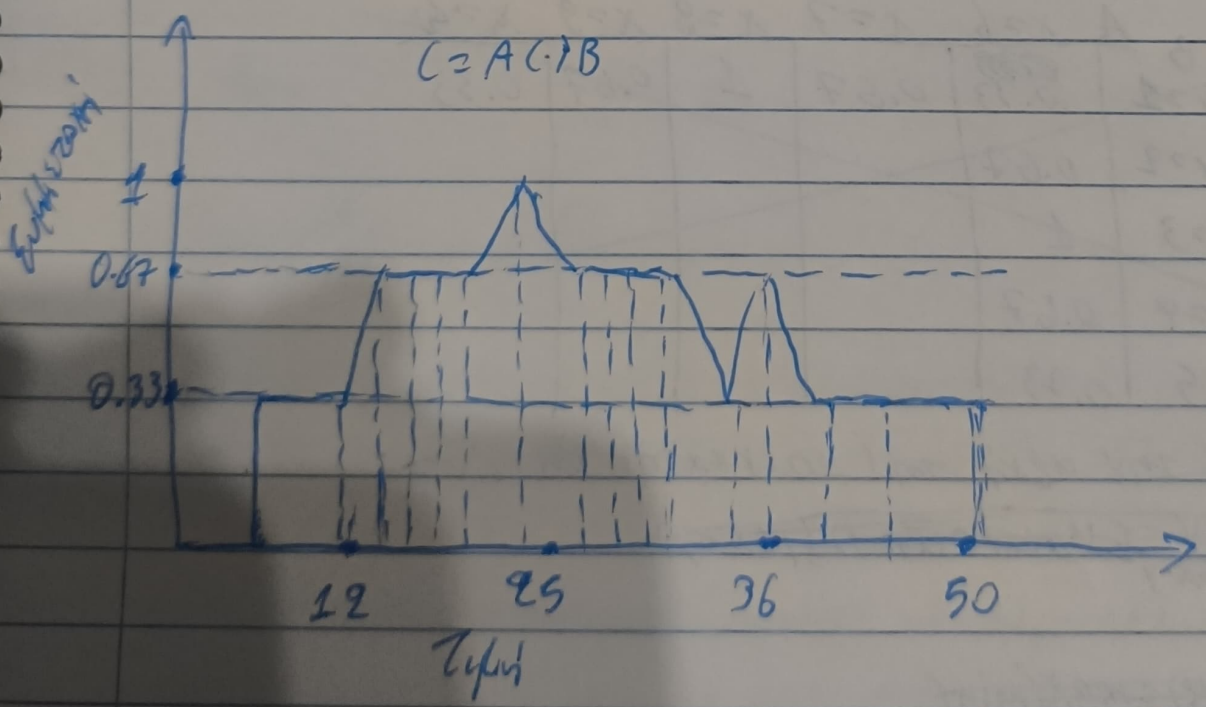
$\mu_{C=A\cdot B}(35) = \min(0.67, 0.33) = \underline{0.33}$

$\mu_{C=A\cdot B}(36) = \min(0.67, 0.67) = 0.67$

$\mu_{C=A\cdot B}(40) = \max(\min(1.00, 0.33), \min(0.33, 0.67)) = \underline{0.33}$

$\mu_{C=A\cdot B}(45) = \min(0.67, 0.33) = \underline{0.33}$

$\mu_{C=A\cdot B}(50) = \min(0.33, 0.33) = \underline{0.33}$



iii) $D = A(-)B$ $\quad \mu_{A-B}(z) = \max\min(\mu_A(x), \mu_B(y))$

$z = x - y$

$z=1$: $x=6, y=5$: $\min(0.33, 0.33) = \underline{0.33}$

$z=2$: $x=7, y=5$: $\min(0.67, 0.33) = 0.33$ } $\max(0.33, 0.33) = \underline{0.33}$

$x=6, y=4$: $\min(0.33, 0.67) = 0.33$

$z=3$: $\max(\min(0.33, 1.00), \min(0.67, 0.67), \min(1.00, 0.33)) = \boxed{0.67}$

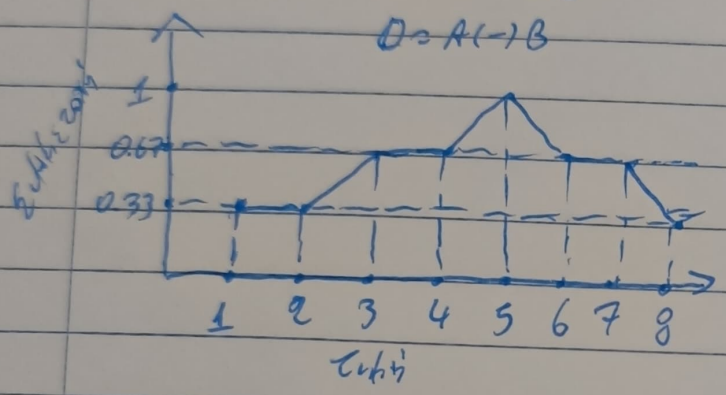
$z=4$: $\max(\min(0.67, 1.00), \min(1.00, 0.67), \min(0.67, 0.33)) = 0.67$

$z=5$: $\max(\min(0.33, 0.33), \min(0.67, 0.67), \min(1.00, 1.00), \min(0.67, 0.67),$
$\min(0.33, 0.33)) = \boxed{1.00}$

$z=6$: $\max(\min(1.00, 0.67), \min(0.67, 1.00), \min(0.33, 0.67)) = 0.67$

$z=7$: $\max(\min(0.67, 0.67), \min(0.33, 1.00)) = 0.67$

$z=8$: $\min(0.33, 0.67) = \underline{0.33}$



$E = A(\div)B$ $\quad \mu_{A/B}(z) = \max\min(\mu_A(x), \mu_B(y))$

$z = x/y$

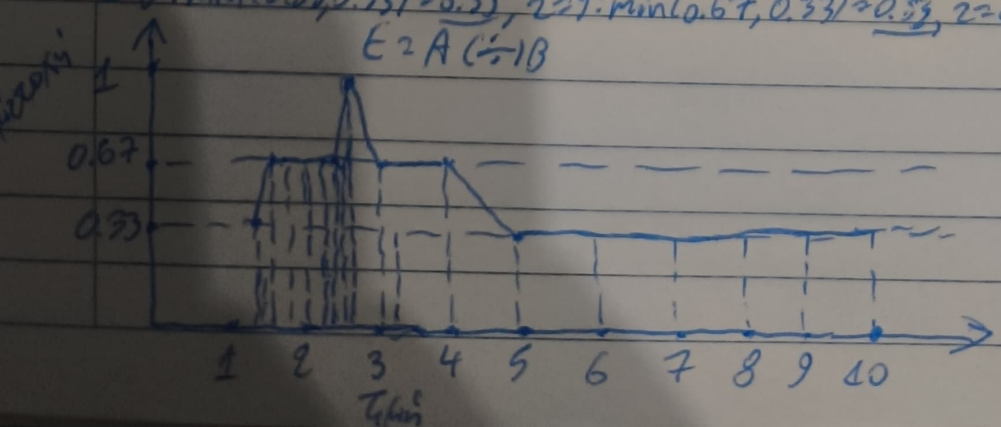
$z=6$: $\min(0.33, 0.33) = \underline{0.33}$, $z=3$: $\max(\min(0.33, 0.67), \min(0.67, 1.00)) = 0.67$

$z=2.33$: $\min(0.67, 1.00) = 0.67$, $z=2$: $\max(\min(1.00, 0.67), \min(0.33, 1.00), \min(0.33, 0.$

$z=1.75$: $\min(0.67, 0.67) = 0.67$, $z=2.67$: $\min(1.00, 1.00) = \boxed{1.00}$ $z=2.25$: $\min(0.67, 0.67) = 0.67$

$z=1.5$: $\min(0.33, 0.67) = \underline{0.33}$, $z=1.6$: $\min(1.00, 0.33) = \underline{0.33}$, $z=1.4$: $\min(0.67, 0.33) = \underline{0.33}$

$z=1.8$: $\min(0.67, 0.33) = \underline{0.33}$, $z=2.5$: $\min(0.33, 0.67) = \underline{0.33}$, $z=3.33$: $\min(0.33, 1.00) = 0.33$

$z=4.0$: $\min(1.00, 0.67) = 0.67$, $z=5.0$: $\min(0.33, 0.67) = \underline{0.33}$, $z=7$: $\min(0.67, 0.33) = \underline{0.33}$

$z=8$: $\min(1.00, 0.33) = \underline{0.33}$, $z=9$: $\min(0.67, 0.33) = \underline{0.33}$, $z=10$: $\min(0.33, 0.33) = \underline{0.33}$